NOVA (**N**eural **O**utput **V**isualization and **A**nalysis) is an R package for analysing Multi-Electrode Array (MEA) neuronal recordings. It transforms raw electrode CSV files into publication-ready figures with a single, reproducible workflow — no programming experience required beyond changing one file path.

MEA experiments record the electrical activity of neuronal networks across dozens of wells, timepoints, treatments and genotypes. The resulting data is high-dimensional: many metrics × many wells × many timepoints. NOVA addresses three core analytical needs:

1. **Dimensionality reduction** — PCA collapses all electrode metrics into interpretable axes, revealing which treatments shift network activity and how genotype modifies the response.
2. **Temporal trajectory analysis** — connecting PCA scores across timepoints reveals the *path* each treatment group takes through network-state space, distinguishing fast responders from slow.
3. **Per-metric visualisation** — bar, box and violin plots for any individual electrode metric (firing rate, burst frequency, synchrony…) across groups and timepoints.

NOVA

Neural Output Visualization

Step	Function	What it does
1. Discover	`discover_mea_structure()`	Scans your data folder, detects experiments, timepoints, well counts and treatment metadata automatically.
2. Process	`process_mea_flexible()`	Reads all CSVs, assigns Treatment/Genotype from filenames, baseline-normalises values, returns tidy long-format data.
3. Analyse	`pca_analysis_enhanced()`	Runs PCA on normalised electrode metrics, computes variance explained, returns scores and loadings.
4. Visualise	`plot_pca_trajectories_general()` `create_mea_heatmaps_enhanced()` `plot_mea_metric()`	Creates trajectory plots, heatmaps and per-metric figures. All functions return ggplot/pheatmap objects and optionally save to disk.

Alex Tudoras • UCSF • alex.tudorasmiravet@ucsf.edu

github.com/atudoras/nova

Installation & Quick Start

Installing NOVA

NOVA is available from GitHub. Install it in R with:

```
# Recommended
remotes::install_github("atudoras/nova")

# Or using devtools
devtools::install_github("atudoras/nova")
```

Tip: If you do not have `remotes` installed yet, run `install.packages("remotes")` first.

CRAN submission is pending. Once accepted, you will be able to install via:

```
install.packages("NOVA")
```

Zero-Code Quick Start

The fastest way to run a full NOVA analysis is the included quickstart script. It requires only one line to be edited:

```
# 1. Open: Example/nova_quickstart.R
# 2. Change this one line to point to your MEA data folder:
DATA_DIR <- "/path/to/your/MEA/data/folder"

# 3. Source the script — all figures save to DATA_DIR/nova_output/
source("Example/nova_quickstart.R")
```

Note: The quickstart expects one subfolder per experiment and one CSV per timepoint inside each subfolder (e.g. baseline.csv, 15min.csv). Folder names can be anything — Plate01, Exp_A, MEA022b.

Loading the Package

```
library(NOVA)
```

All functions are available after loading. No additional setup is needed.

Data Requirements & Directory Structure

Expected Folder Layout

NOVA expects one subfolder per experiment and one CSV file per timepoint inside each subfolder. Folder names can be anything that makes sense for your experiment — NOVA uses them as labels in figures.

```
Data/
Experiment_A/
baseline.csv
15min.csv
30min.csv
1h.csv
2h.csv
Experiment_B/
baseline.csv
...

# Folder names like MEA022b, Plate01, Ctrl_Run3 all work equally well.
```

Rules:

- Each experiment is a subfolder — any name is accepted.
- Each timepoint is a separate CSV file — the filename becomes the timepoint label.
- Folder names and CSV names are used verbatim as labels in all figures.
- If your folder names follow an unusual pattern, pass a matching regex via `experiment_pattern` in `discover_mea_structure()`.

CSV Format

NOVA auto-detects the metadata region using a label scan — no hardcoded row numbers. The expected format is:

```
# Rows 1–120: plate map / instrument header (auto-skipped)
# Row 121+: data region
#
# Column A: Well label (A1, B1, C1...)
# Column B: Treatment assignment <- detected by label scan
# Column C: Genotype assignment <- detected by label scan
# Columns D+: one column per MEA metric (Firing Rate, etc.)
```

Tip: If your CSVs do not have Treatment/Genotype columns, NOVA will still process the data and use the experiment folder name as the group label. You can assign metadata manually via the config argument.

Supported Timepoint Naming

NOVA recognises these timepoint naming conventions and auto-sorts them chronologically:

```
# Supported patterns (case-insensitive):
baseline, bl, base
15min, 15m, 15
30min, 30m, 30
1h, 1hour, 60min
1h30min, 1h30m, 90min
2h, 2hour, 120min
1day, 24h, 1d

# Custom names also work — just specify timepoint_order manually
```

Core Workflow: Step by Step

Step 1 — Discover Your Data

Start by scanning the data folder. This builds the structure map that all downstream functions use.

```
library(NOVA)

structure <- discover_mea_structure(
main_dir = "/path/to/MEA_Data",
verbose = TRUE # prints a summary table
)

# Returns a list with:
# $experiments — experiment names found
# $timepoints — timepoints per experiment
# $n_wells — well counts
# $treatments — treatment labels detected
```

Step 2 — Process the Data

Load all CSVs, assign metadata, and normalise to baseline.

```
result <- process_mea_flexible(
main_dir = "/path/to/MEA_Data",
grouping_variables = c("Treatment", "Genotype"),
baseline_timepoint = "baseline", # NULL = no normalisation
exclude_std_variables = TRUE, # remove SD columns
verbose = TRUE
)

# result$normalized_data — long-format tidy data frame
# result$metadata — well-level treatment/genotype table
# result$summary — processing summary
```

Step 3 — Run PCA

Reduce the multi-metric data to principal components.

```
pca_out <- pca_analysis_enhanced(
processing_result = result,
n_components = 10, # number of PCs to retain
variance_cutoff = 0.95, # retain PCs up to 95% variance
scale_data = TRUE, # z-score before PCA (recommended)
impute = TRUE, # impute missing wells
verbose = TRUE
)

# pca_out$plot_data — PC scores with Treatment/Genotype/Timepoint
# pca_out$pca_result — prcomp object
# pca_out$variance — variance explained per PC
```

Tip: PC1 and PC2 together typically explain 50–70% of variance in MEA data. If less than 50%, consider filtering out low-variance metrics with min_var before running PCA.

PCA Trajectory Analysis

Trajectory plots connect each group's PCA centroid across timepoints, revealing the *path* each treatment takes through network-state space. NOVA generates individual-well trajectories, group averages with error bars, and colour-coded multi-group overlays.

Basic Trajectory Plot

```
plots <- plot_pca_trajectories_general(
pca_results = pca_out,
trajectory_grouping = c("Treatment", "Genotype"),
timepoint_var = "Timepoint",
timepoint_order = c("baseline", "15min", "30min", "1h", "1h30min", "2h"),
line_size = 2,
alpha = 0.8,
return_list = TRUE,
save_plots = FALSE
)

# Access individual plots:
plots$combined_avg # all groups, group averages
plots$combined_all # all individual wells
plots$KA_avg # KA group average only
```

Color by Treatment (New in v0.1.1)

The new `color_by` parameter colours each trajectory by a single variable rather than a group combination. This is ideal for README figures and presentations.

```
plots <- plot_pca_trajectories_general(
pca_results = pca_out,
color_by = "Treatment", # NEW: colour by Treatment only
trajectory_grouping = c("Treatment"),
timepoint_order = c("baseline", "15min", "30min",
"1h", "1h30min", "2h")
)
```

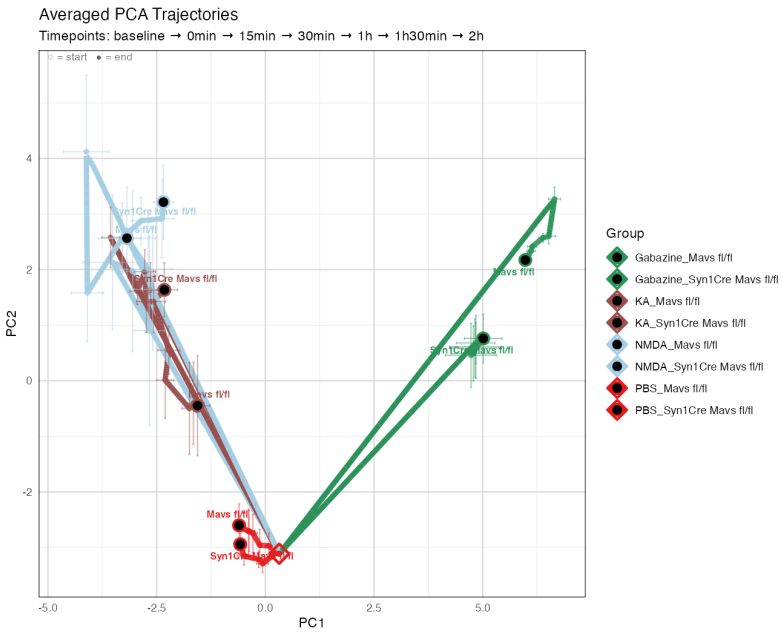

Figure 5.2 — Trajectories coloured by Treatment. Genotype labels appear at the 2 h endpoint via ggrepel.

Saving Trajectory Plots

```
plot_pca_trajectories_general(
pca_results = pca_out,
save_plots = TRUE,
output_dir = "/path/to/output/",
plot_prefix = "my_experiment",
width = 10,
height = 8,
dpi = 300
)
```

Trajectory Plot Parameters

Parameter	Default	Description
`pca_results`	`required`	PCA output from pca_analysis_enhanced(), or a data frame with PC1/PC2/Timepoint columns.
`color_by`	`"group"`	Variable to colour trajectories by. Options: "group" (Treatment+Genotype combination), "Treatment", or any column in plot_data.
`trajectory_grouping`	`NULL`	Character vector of grouping variables. NULL = auto-detect Treatment and Genotype columns.
`timepoint_order`	`NULL`	Character vector giving chronological order of timepoints. NULL = auto-detect.
`line_size`	`2`	Line width for trajectory paths.

Parameter	Default	Description
`alpha`	`0.7`	Transparency for individual-well trajectories (0=transparent, 1=opaque).
`smooth_lines`	`FALSE`	Apply spline smoothing to trajectory curves.
`color_palette`	`NULL`	Named character vector of hex colours. NULL = auto-generate from built-in scientific palette.
`save_plots`	`FALSE`	If TRUE, save all plots to output_dir as PNG files.
`dpi`	`150`	Resolution for saved plots. Use 300 for publication.

Heatmap Analysis

NOVA's heatmap function renders the multi-metric MEA data as a clustered colour grid. Three modes are available: by Treatment, by Genotype, and the new **combination** mode (Treatment x Genotype) added in v0.1.1.

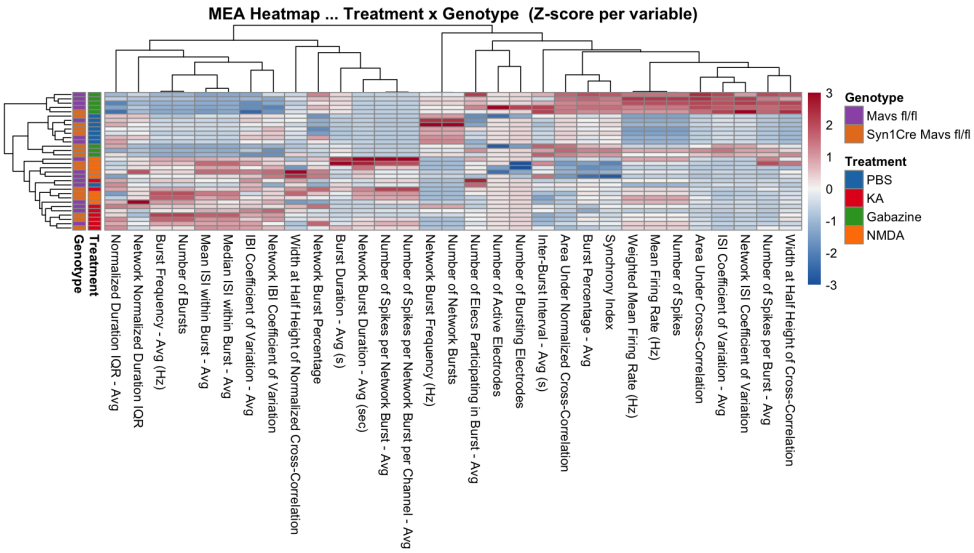

Figure 6.1 — Treatment x Genotype combination heatmap with Z-score scaling (blue = below mean, red = above mean, capped at +-3 SD). Dual annotation strips show Treatment (top) and Genotype (right).

Combination Heatmap (Treatment x Genotype)

The combination heatmap is the recommended default — it shows both experimental variables simultaneously and uses Z-score normalisation to make all metrics visually comparable regardless of scale.

```
heatmaps <- create_mea_heatmaps_enhanced(
processing_result = result,
split_by = "combination", # Treatment x Genotype
scale_method = "z_score",
cluster_rows = TRUE,
cluster_cols = TRUE,
save_plots = FALSE
)

# Access:
heatmaps$combination_result # pheatmap object
```

Treatment-Only and Genotype-Only Heatmaps

```
# Treatment only
create_mea_heatmaps_enhanced(
processing_result = result,
split_by = "Treatment"
)

# Genotype only
create_mea_heatmaps_enhanced(
processing_result = result,
split_by = "Genotype"
)
```

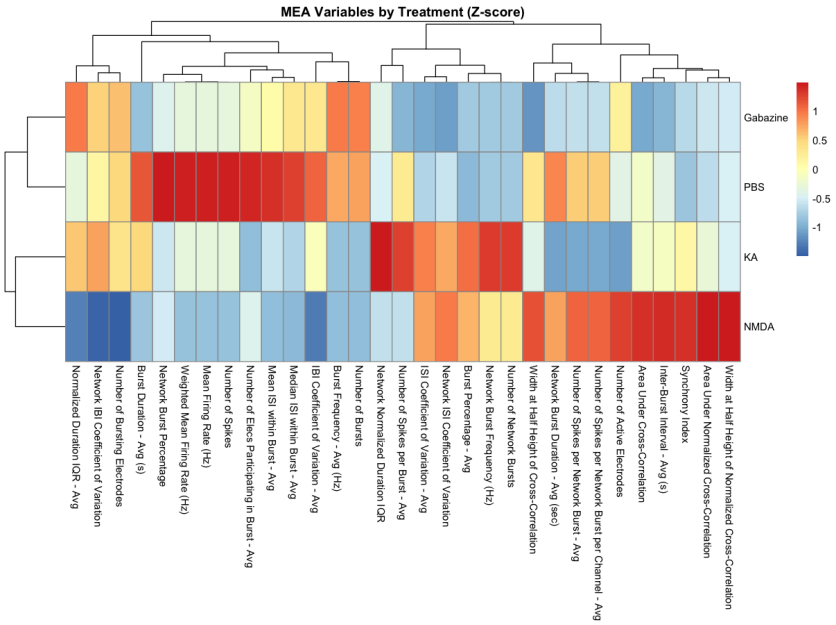

Figure 6.2 — Treatment-only heatmap, wells averaged per treatment group.

Filtering Data Before Heatmap

```
# Show only KA and PBS at timepoints 1h and 2h
create_mea_heatmaps_enhanced(
processing_result = result,
filter_treatments = c("KA", "PBS"),
filter_timepoints = c("1h", "2h"),
split_by = "combination"
)
```

Heatmap Parameters

Parameter	Default	Description
`split_by`	`NULL`	How to split wells into heatmap annotation columns. Options: NULL, "Treatment", "Genotype", "combination".

Parameter	Default	Description
`scale_method`	`"z_score"`	Normalisation before colour mapping. Options: "z_score", "none", "row", "column".
`cluster_rows`	`TRUE`	Hierarchical clustering of metrics (rows).
`cluster_cols`	`TRUE`	Hierarchical clustering of wells (columns).
`filter_timepoints`	`NULL`	Character vector — include only these timepoints. NULL = all.
`filter_treatments`	`NULL`	Character vector — include only these treatments. NULL = all.
`filter_genotypes`	`NULL`	Character vector — include only these genotypes. NULL = all.
`use_raw`	`FALSE`	If TRUE, use raw electrode values instead of normalised values.
`fontsize`	`10`	Base font size for row/column labels.
`dpi`	`300`	Resolution for saved plots.

Per-Metric Visualisation

While PCA and heatmaps show the whole dataset, `plot_mea_metric()` focuses on a *single* electrode metric at a time — letting you inspect firing rate, burst frequency, synchrony, or any other metric in detail with bar, box, violin or line charts.

Bar Chart with SEM Error Bars

```
plot_mea_metric(
data = result$normalized_data,
metric = "Firing Rate",
x_var = "Timepoint",
group_by = "Treatment",
plot_type = "bar",
error_type = "sem", # "sem", "sd", or "ci95"
show_points = TRUE,
point_alpha = 0.5
)
```

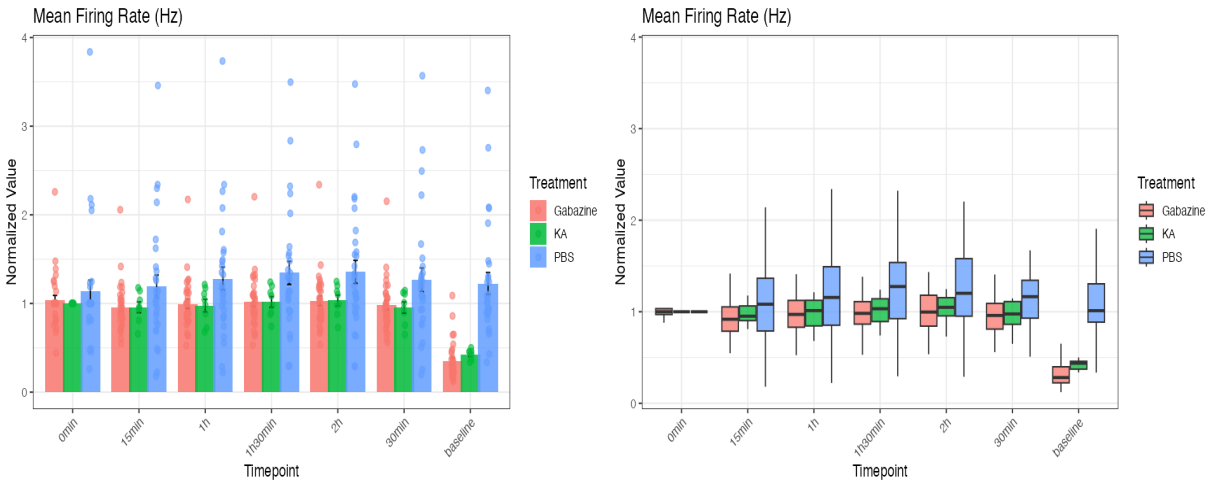

Figure 7.1 — Bar chart with SEM (left) and box plot (right) for Firing Rate across treatments over time.

Box and Violin Plots

```
# Box plot
plot_mea_metric(data=result$normalized_data, metric="Firing Rate",
plot_type="box", group_by="Treatment")

# Violin plot with individual points
plot_mea_metric(data=result$normalized_data, metric="Burst Frequency",
plot_type="violin", show_points=TRUE, point_alpha=0.4)
```

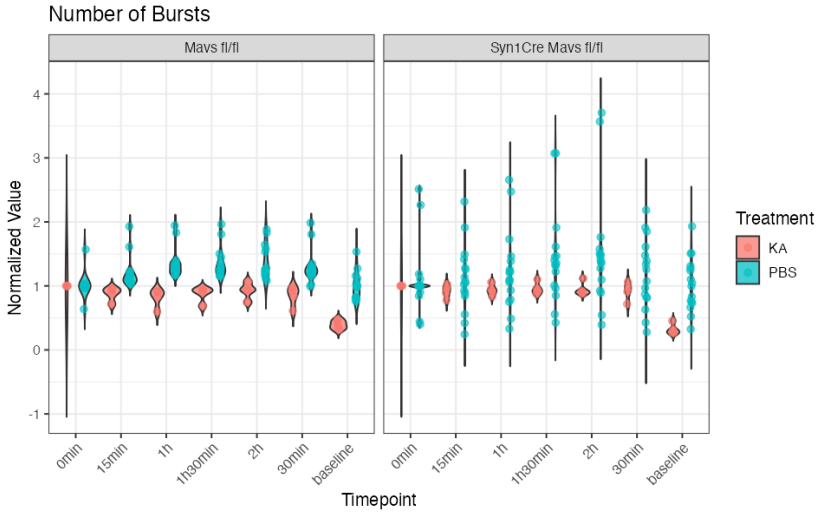

Figure 7.2 — Violin plot of Burst Frequency across treatment groups, with individual well data points overlaid.

Faceting by Genotype

```
# Split into panels by Genotype
plot_mea_metric(
data = result$normalized_data,
metric = "Synchrony Index",
x_var = "Timepoint",
group_by = "Treatment",
facet_by = "Genotype", # one panel per genotype
plot_type = "bar"
)
```

plot_mea_metric Parameters

Parameter	Default	Description
`data`	`required`	Long-format data frame — typically result$normalized_data from process_mea_flexible().
`metric`	`required`	Exact string matching a value in the Variable column (e.g. "Firing Rate", "Burst Frequency").
`x_var`	`"Timepoint"`	Column to use as the x-axis.
`group_by`	`"Treatment"`	Grouping variable for colour and fill.
`facet_by`	`NULL`	Column to facet (split into separate panels). NULL = no faceting.
`plot_type`	`"bar"`	Chart type: "bar", "box", "violin", or "line".
`error_type`	`"sem"`	Error bar type: "sem" (standard error), "sd" (standard deviation), "ci95" (95% CI).

Parameter	Default	Description
`show_points`	`TRUE`	Overlay individual data points on bar/violin plots.
`filter_treatments`	`NULL`	Subset of treatments to include. NULL = all.
`filter_timepoints`	`NULL`	Subset of timepoints to include. NULL = all.

Customising Figures

Custom Colour Palettes

All NOVA plotting functions accept a `color_palette` argument — a named character vector mapping group labels to hex colours.

```
# Define your own palette
my_palette <- c(
"PBS" = "#6B7280", # gray — control
"KA" = "#EF4444", # red — seizure
"Gabazine" = "#3B82F6", # blue — disinhibition
"NMDA" = "#10B981" # green — excitatory
)

plot_pca_trajectories_general(
pca_results = pca_out,
color_palette = my_palette
)
```

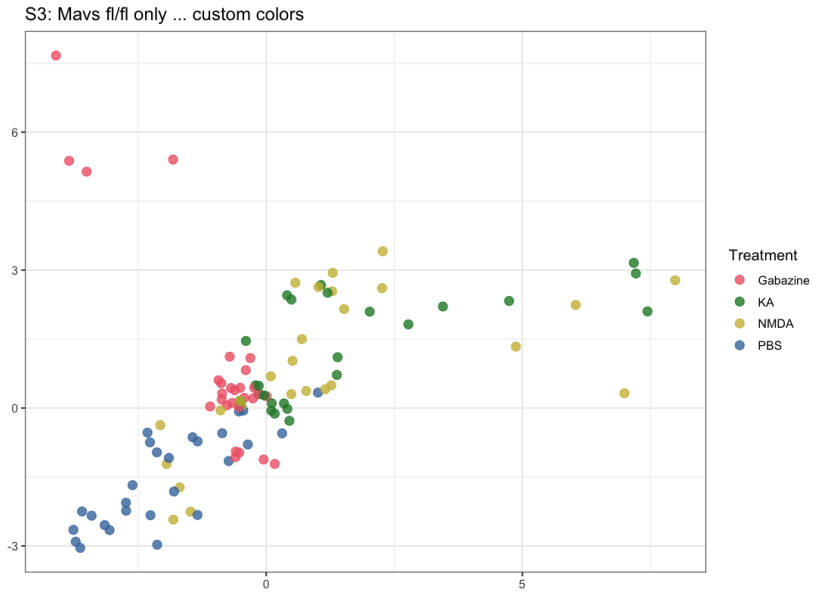

Figure 8.1 — PCA scatter with a custom four-colour palette.

Saving All Figures at Once

```
# Save trajectory plots at publication quality
plot_pca_trajectories_general(
pca_results = pca_out,
save_plots = TRUE,
output_dir = "figures/trajectories/",
dpi = 300,
width = 10,
height = 8
)

# Save heatmaps
create_mea_heatmaps_enhanced(
processing_result = result,
save_plots = TRUE,
output_dir = "figures/heatmaps/",
plot_format = "png", # or "pdf", "svg"
dpi = 300
)
```

Changing PCA Components

```
# Plot PC3 vs PC4 instead of PC1 vs PC2
plot_pca_trajectories_general(
pca_results = pca_out,
pc_x = "PC3",
pc_y = "PC4"
)
```

Complete Function Reference

All exported NOVA functions are listed below with their full signatures. Access in-line help with `?function_name`.

discover_mea_structure()

Parameter	Default	Description
`main_dir`	`required`	Path to the folder containing MEA experiment subfolders.
`experiment_pattern`	`".*"`	Regex matching experiment subfolder names. Default matches any folder; narrow it if non-experiment folders are present (e.g. '"Plate\\d+"').
`file_pattern`	`"\\.csv$"`	Regex matching CSV filenames to include.
`verbose`	`TRUE`	Print a discovery summary table.

process_mea_flexible()

Parameter	Default	Description
`main_dir`	`required`	Path to MEA data folder.
`selected_experiments`	`NULL`	Character vector — process only these experiments. NULL = all.
`selected_timepoints`	`NULL`	Character vector — include only these timepoints. NULL = all.
`grouping_variables`	`c("Treatment","Genotype")`	Column names to use as grouping factors.
`baseline_timepoint`	`NULL`	Timepoint name to use as baseline (values divided by baseline mean). NULL = no normalisation.
`exclude_std_variables`	`TRUE`	Drop standard-deviation metric columns.
`verbose`	`TRUE`	Print processing log.

pca_analysis_enhanced()

Parameter	Default	Description
processing_result	NULL	Output of process_mea_flexible(). Required if normalized_data is NULL.
n_components	10	Number of principal components to compute.
variance_cutoff	0.95	Retain components explaining this cumulative variance.
scale_data	TRUE	Z-score metrics before PCA (strongly recommended).
impute	TRUE	Impute missing wells using column mean.
min_var	NULL	Minimum variance threshold; metrics below this are excluded.
value_column	"Normalized_Value"	Column containing metric values.
verbose	TRUE	Print PCA summary.

plot_pca_trajectories_general()

Parameter	Default	Description
pca_results	required	PCA output object or data frame with PC columns.
color_by	"group"	Colour variable: "group", "Treatment", or any column name.
trajectory_grouping	NULL	Grouping variables for trajectory lines. NULL = auto-detect.
timepoint_var	"Timepoint"	Column identifying timepoints.
timepoint_order	NULL	Chronological order of timepoints. NULL = auto-sort.
individual_var	"Experiment"	Variable identifying individual wells/experiments.
point_size	3	Size of timepoint dots.
line_size	2	Line width for trajectory paths.
alpha	0.7	Transparency for individual traces.
smooth_lines	FALSE	Apply spline smoothing.
color_palette	NULL	Named colour vector. NULL = built-in palette.
save_plots	FALSE	Save to output_dir.
dpi	150	Resolution for saved files.
return_list	TRUE	Return list of all plot objects.

create_mea_heatmaps_enhanced()

Parameter	Default	Description
processing_result	NULL	Output of process_mea_flexible().
split_by	NULL	NULL, "Treatment", "Genotype", or "combination".
scale_method	"z_score"	"z_score", "none", "row", or "column".
cluster_rows	TRUE	Cluster metrics (rows).
cluster_cols	TRUE	Cluster wells (columns).
filter_timepoints	NULL	Include only these timepoints.
filter_treatments	NULL	Include only these treatments.
filter_genotypes	NULL	Include only these genotypes.
use_raw	FALSE	Use raw values instead of normalised values.
quality_threshold	0.8	Minimum data completeness (0–1) for a well to be included.
min_observations	3	Minimum number of observations per well.
save_plots	FALSE	Save to output_dir.
dpi	300	Resolution.

plot_mea_metric()

Parameter	Default	Description
data	required	Long-format data frame (result$normalized_data).
metric	required	Metric name string matching Variable column.
x_var	"Timepoint"	X-axis variable.
group_by	"Treatment"	Colour/fill grouping variable.
facet_by	NULL	Faceting variable. NULL = no facets.
plot_type	"bar"	"bar", "box", "violin", or "line".
error_type	"sem"	"sem", "sd", or "ci95".
show_points	TRUE	Overlay individual data points.
point_alpha	0.6	Transparency for data points.
colors	NULL	Named colour vector. NULL = auto.

Troubleshooting

discover_mea_structure() finds 0 experiments

Check that your data folder contains subfolders — one per experiment. Folder names can be anything. If only some subfolders are being missed, use experiment_pattern to restrict which folders NOVA scans, e.g. experiment_pattern = "Plate\\d+" to match Plate01, Plate02, etc.

CSV files are read but all values are NA

NOVA auto-detects the data region by scanning for the label row. If your CSV header is at a non-standard position, the scan may fail. Check your CSV structure and ensure the column containing well labels (A1, B1...) is present.

PCA plot shows only one cluster / no separation

This usually means baseline normalisation collapsed the variance. Try setting baseline_timepoint = NULL in process_mea_flexible() to use raw values, or check that your baseline CSV has sufficient coverage.

Trajectory lines look random / cross unexpectedly

Check timepoint_order — if timepoints are not in the correct chronological order, trajectories will zigzag. Always specify timepoint_order explicitly when using non-standard timepoint names.

Heatmap shows only one colour (all red or all blue)

This indicates extreme outlier wells are dominating the colour scale. The combination heatmap already caps Z-scores at +-3 SD, but if using scale_method = "none", try switching to "z_score".

Error: package ggrepel not found

Install ggrepel: install.packages("ggrepel"). It is required for trajectory end-point labels.

Saved figures are blurry

Increase dpi to 300 (publication standard) in save_plots calls. The default dpi = 150 is suitable for screen preview only.

For additional help, open an issue at **github.com/atudoras/nova/issues** or email alex.tudorasmiravet@ucsf.edu.